MTH 451/551, Fall 2026
Quiz #3, Tue., Nov. 10, 2026

Solutions

1. Let $x_0 = 0$, $x_1 = 1$, $x_2 = 2$, and let $y_0 = 1$, $y_1 = 3$, $y_2 = 7$. Write down the Lagrange polynomials ℓ_0, ℓ_1, ℓ_2 associated with the points $\{x_0, x_1, x_2\}$. Write the formula for the interpolating polynomial $p \in \mathbb{P}_2$ satisfying $p(x_j) = y_j$ for each j. (You do not need to expand p.)

Solution. Using $\ell_j(x) = \prod_{k \neq j} \frac{x - x_k}{x_j - x_k}$,

$$\ell_0(x) = \frac{(x-1)(x-2)}{(0-1)(0-2)} = \frac{(x-1)(x-2)}{2},$$
$$\ell_1(x) = \frac{(x-0)(x-2)}{(1-0)(1-2)} = -x(x-2),$$
$$\ell_2(x) = \frac{(x-0)(x-1)}{(2-0)(2-1)} = \frac{x(x-1)}{2}.$$

The interpolating polynomial is

$$p(x) = y_0\ell_0(x) + y_1\ell_1(x) + y_2\ell_2(x) = \frac{(x-1)(x-2)}{2} - 3x(x-2) + \frac{7x(x-1)}{2}.$$

(Expanding, $p(x) = x^2 + x + 1$, which indeed gives $p(0) = 1$, $p(1) = 3$, $p(2) = 7$).

2. (a) Which of the following are linear functionals on $\mathbb{P}_n$? Give brief justifications.

$$\phi(p) = p(0) + p'(1), \qquad \psi(p) = p(0) + 1, \qquad \chi(p) = \big(p(0)\big)^2.$$

(b) Let $\phi_0(p) = p(0)$, $\phi_1(p) = p'(0)$, and $\phi_2(p) = p(1)$ for $p \in \mathbb{P}_2$. Show that the only $p \in \mathbb{P}_2$ with $\phi_0(p) = \phi_1(p) = \phi_2(p) = 0$ is $p \equiv 0$. (**Hint:** what are the roots, with multiplicity, of such a p?) Do these functionals specify a unique polynomial?

Solution.

(a)

- ϕ is linear by linearity of evaluation and linearity of the derivative.
- ψ is not linear, since $\phi(0) = 1 \neq 0$.
- χ is not linear, since taking $p \equiv 1$, we have $\chi(2p) = 4$, but $2\chi(p) = 2$.

(b) Suppose $p \in \mathbb{P}_2$ satisfies $p(0) = p'(0) = 0$ and $p(1) = 0$. Since $p(0) = p'(0) = 0$, $r = 0$ is a root of p of multiplicity at least 2, and since $p(1) = 0$, $r = 1$ is another root. So p has at least 3 roots, counting multiplicities. Since $\deg(p) \leq 2$, the Fundamental Theorem of Algebra implies $p \equiv 0$.

(Alternatively, write $p(x) = a_0 + a_1x + a_2x^2$. Then $p(0) = a_0 = 0$ and $p'(0) = a_1 = 0$, so $p(1) = a_2 = 0$.)

By Lemma 3.4, for any numbers y_0, y_1, y_2, there is a unique $p \in \mathbb{P}_2$ with $p(0) = y_0$, $p'(0) = y_1$, and $p(1) = y_2$. (Explicitly, $p(x) = y_0 + y_1x + (y_2 - y_0 - y_1)x^2$.)